

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Ignition Coil Tester Kit, Part Number 4918343

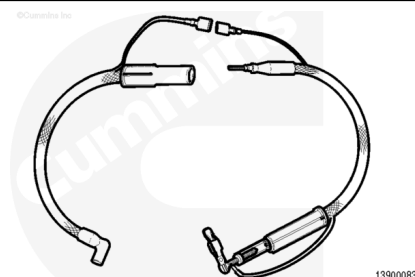
Additional Service Items

- Harness extender.

Initial Check

⚠ WARNING ⚠

The ignition system produces high voltage. To reduce the possibility of personal injury from electrical shock, do not touch any ignition wires or components while the engine is operating. Always use suitably insulated tools.



⚠ WARNING ⚠

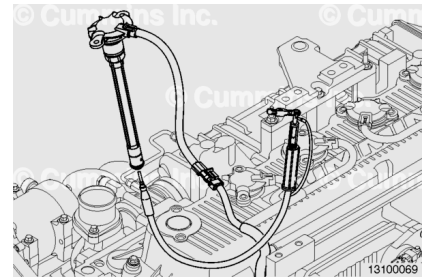
The engine must not be running and the ignition or key switch must be in the OFF position when installing or removing the ignition generator. To reduce the possibility of personal injury or equipment damage, do not touch the pins on the ignition generator when the engine is operating. Always use suitably insulated tools.

⚠ CAUTION ⚠

The alligator clip must be grounded for the tool to function. If the tool is not grounded, arcing can result from the high voltage being applied to the alligator clip.

⚠ CAUTION ⚠

Do not attempt to adjust the ignition coil tester. It is preset and not adjustable. An attempt to adjust can result in tool damage.



Use a harness extender.

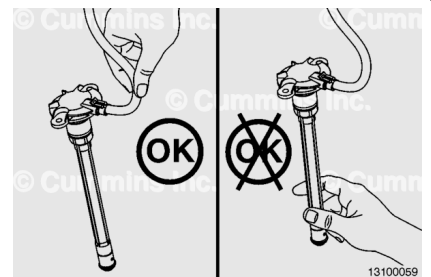
Remove the ignition coil.

Connect the ignition harness to the harness extender, then connect the harness extender to the ignition coil. Insert the bare thread of the ignition coil tester included in the Ignition Coil Tester Kit, Part Number 4918343, into the end of the ignition coil.

Attach the alligator clip to a suitable ground.

Route the ignition coil tester so that it does **not** touch any metallic surface on the engine or arcing may occur.

Holding the connector coil by the ignition harness is acceptable.



Crank the engine or start the engine and let it idle.

Observe the spark gap inside the observation window. If a steady spark is observed, the ignition coil is acceptable; if no spark is observed, replace the ignition coil and test again.

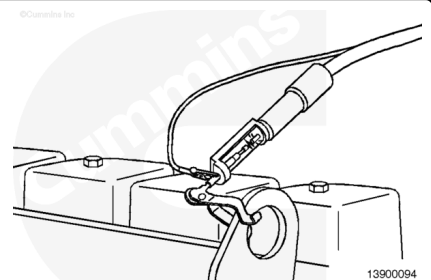
Note : It may take several revolutions of the engine before a spark is observed.

Stop the engine.

Remove the ignition coil tester from the ignition coil and disconnect the alligator clip.

Install the ignition coil.

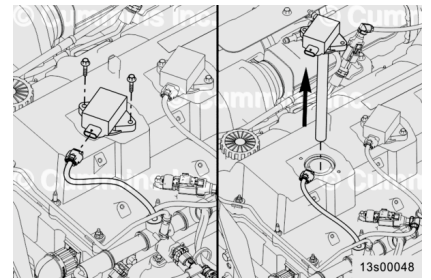
Repeat for the remaining ignition coils.



Remove

⚠ WARNING ⚠

The ignition system produces high voltage. To reduce the possibility of personal injury from electrical shock, do not touch any ignition wires or components while the engine is operating. Always use suitably insulated tools.



Disconnect the ignition wiring harness from the ignition coil.

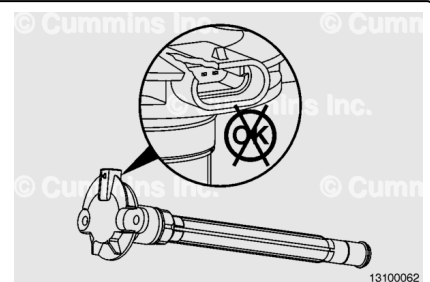
Remove the capscrews.

Remove the ignition coil by pulling directly upward.

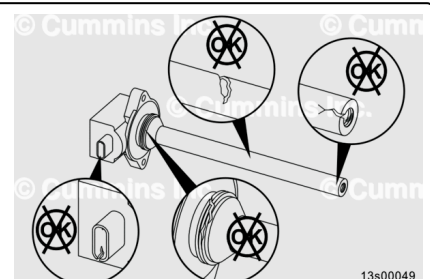
Inspect for Reuse

Inspect the ignition coil body for cracks, tears, broken connector pins, and other signs of damage.

Replace the ignition coil if damage is found.



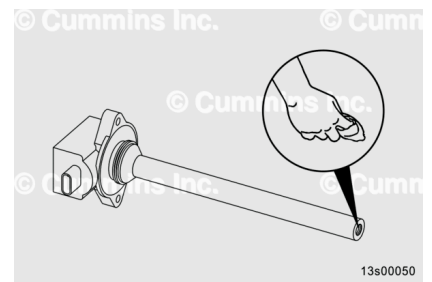
Refer to Service Bulletin, Ignition Coil Malfunctions, Bulletin 4383630 (/qs3/pubsys2/xml/en/bulletin/4383630.html), to determine if the coil is reusable.



Install

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

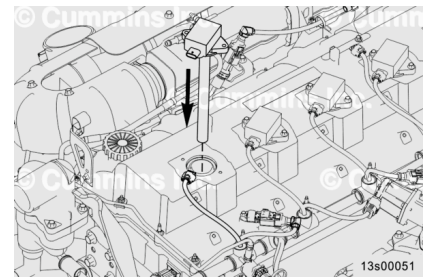


Apply a pea-sized amount of Cummins Inc. approved dielectric grease to the spark plug end of the extension using a cotton swab.

Use a cotton swab and isopropyl alcohol to remove any excess dielectric grease from the spark plug end of the coil extension.

Note : New coil extensions are pre-greased on the spark plug end of the coil extension and do not require any additional lubrication prior to installation.

Press downward on the ignition coil until the grommet seal contacts the valve cover, then align the coil mounting holes with the mounting holes in the valve cover.



Install the capscrews.

Tighten the capscrews.

Torque Value: 10 n•m [89 in-lb]

Connect the ignition wiring harness to the ignition coil.

